CIE AS PHYSICS (9702) • QUESTION PAPER

# 1.1 Physical Quantities

**Syllabus Section:** 1. Physical Quantities & Units **Paper Type:** Structured Questions (Paper 2) **Total:** 8 Questions (59 Marks)

## Question 1 Easy [4 marks]

**Part (a)** [1 mark]

State what is meant by a physical quantity.

**Part (b)** [1 mark]

State **one** physical quantity that can be represented by the symbol $\rho$.

**Part (c)** [1 mark]

State **one** physical quantity that can be represented by the capital letter $P$ or the lowercase letter $p$.

**Part (d)** [1 mark]

Explain how the quantities represented by $\rho$, $p$, or $P$ can be distinguished from each other.

## Question 2 Easy [6 marks]

### Part (a) [3 marks]

Complete Table 1.1 to show the name and symbol of each physical quantity and the corresponding unit.

**Table 1.1**

| physical quantity | symbol | unit |
|---|---|---|
| velocity | $v$ | m $s^{-1}$ |
| | $W$ | N m |
| stress | $\sigma$ | |
| | | kg $m^{-3}$ |

### Part (b) [3 marks]

Calculate a reasonable estimate for the work done by a person lifting a dining room chair 1.0 m above the ground.

## Question 3 Easy [9 marks]

### Part (a) [3 marks]

A physics student is investigating the properties of materials. She has a stack of 500 sheets of paper and needs to determine the thickness of a single sheet.

(i) Describe a method she could use to find the thickness of one sheet of paper without using a micrometer.

[2]

(ii) State an appropriate unit for the thickness of one sheet of paper.

[1]

### Part (b) [3 marks]

The student then investigates simple harmonic motion using a mass-spring system. She wishes to determine the period of oscillation of the system.

Describe a method the student could use to obtain an accurate value for the period of one complete oscillation.

### Part (c) [1 mark]

During a break in her experiments, the student scrunches up a sheet of paper and throws it horizontally from shoulder height across the laboratory.

Estimate the horizontal distance travelled by the paper ball before it hits the floor.

**Part (d)** **[2 marks]**

The student then throws a tennis ball of the same size using the same technique from the same height.

State whether the tennis ball travels a greater horizontal distance than the paper ball. Explain your answer.

## Question 4 Medium [12 marks]

**Part (a)** [2 marks]

Estimate the speed of an apple as it hits the ground after falling from a tree.

**Part (b)** [2 marks]

Estimate the number of kilowatt-hours used by a lightbulb if switched on for a whole day.

**Part (c)** [4 marks]

Estimate the density, in g $cm^{-3}$, of the head of an adult person.

**Part (d)** [4 marks]

The radius of the Earth is approximately $6.4 \times 10^6$ m.

Estimate the number of people it would take to circle the Earth holding hands.

## Question 5 Medium [12 marks]

### Part (a) [3 marks]

X–ray pulsars are detected by X–ray telescopes on a satellite in low Earth orbit 2000 km above the surface of the Earth.

Calculate the number of cubic millimetres ($mm^3$) in a volume of 2000 $km^3$.

### Part (b) [3 marks]

X–rays from a pulsar travel at the speed of light and are detected on Earth with a wavelength of 8.0 nm.

Calculate the frequency of the X–rays. Give your answer to an appropriate number of significant figures.

### Part (c) [3 marks]

X–rays from the nearest pulsar PSR J0109–1431 take 8.82 Gs to travel to Earth.

Calculate the number of oscillations of the X–rays from the pulsar to the surface of the Earth. Give your answer to an appropriate number of significant figures.

**Part (d)** **[3 marks]**

Show the distance from Earth to PSR J0109–1431 is around 280 light years.

## Question 6 Medium [8 marks]

### Part (a) [1 mark]

Define the term *physical quantity*.

### Part (b) [3 marks]

Physical quantities have symbols and units. Indicate, with a tick (✓), which of the following are **not** valid physical quantities.

| | |
|---|---|
| kinetic energy $E_k$ measured in kg $m^2$ $s^{-2}$ | ☐ |
| pressure $P$ measured in N $kg^{-1}$ | ☐ |
| Young modulus $E$ measured in Pa | ☐ |
| frequency $f$ measured in s | ☐ |
| specific heat capacity $c$ measured in J $kg^{-1}$ | ☐ |
| weight $W$ measured in kg m $s^{-2}$ | ☐ |

### Part (c) [4 marks]

Complete the table providing reasonable estimates for some physical quantities.

| quantity | estimated magnitude |
|---|---|

| | |
|---|---|
| Weight of an adult human | |
| | 3600 s |
| Wavelength of green light | |
| | 20 °C |

## Question 7 Hard [4 marks]

### Part (a) [4 marks]

Complete Table 1.1 to show the SI base unit of each physical quantity and a reasonable estimate of the order of magnitude.

**Table 1.1**

| physical quantity | SI base unit | order of magnitude |
|---|---|---|
| acceleration of free-fall on Earth ($g$) | $m\ s^{-2}$ | |
| Stefan-Boltzmann constant ($\sigma$) | | $10^{-7}$ |
| speed of a $\beta$ particle | | |
| specific heat capacity of water ($c$) | | |

## Question 8 Hard [4 marks]

### Part (a) [4 marks]

Complete Table 2.1 to show a reasonable estimate of the order of magnitude of each physical quantity and the corresponding unit.

**Table 2.1**

| quantity | order of magnitude | unit |
|---|---|---|
| mass of a car | | kg |
| thickness of a sheet of paper | | mm |
| density of air<br>(at room temperature and pressure) | | $g\ cm^{-3}$ |
| time between two heartbeats | | s |